Maxwell's Wheel

A wheel consists of a uniform cylindrical disk of mass $M = 0.40\,\mathrm{kg}$ and radius $R = 6.0\,\mathrm{cm}$ with an axle of negligible mass and radius $r = 3.0\,\mathrm{mm}$. The wheel is suspended from the ceiling on two identical light threads of length $L \gg R$ which are attached to the axle symmetrically (Figure 1). By slowly turning the wheel so that the threads wind around the axle, we raise the disk's centre of mass by $H = 1.0\,\mathrm{m}$, after which the wheel is let go. The wheel descends until it reaches its lowest position and the threads are fully unwound. Then, the threads begin to wind in the opposite direction, and the wheel starts to rise.

(a) Find the angular velocity of the wheel ω after the wheel's centre of mass has covered a distance s in its downwards motion.

(b) Find the kinetic energy E_T of the translational motion of the wheel after it has covered a distance $s = 0.50\,\mathrm{m}$. Find the ratio between E_T and the other types of energy that are relevant to the problem.

(c) Find the tension T in each string as the wheel moves downwards.

In the side view of the wheel (Figure 2), denote the centre of mass by S, the point of contact between the axle and the threads by A, and the suspension point by P. Let $\varphi \in [0; \pi]$ be the angle that the line SA makes with the horizon as the wheel turns around near its lowest point.

(d) Using approximations so that the problem becomes analytically tractable, find the angular velocity of the wheel ω as a function of φ when the wheel turns around. Introduce an appropriate Cartesian frame and write down expressions for the centre of mass' coordinates x and y in terms of φ. Do the same for its velocity components v_x and v_y. Sketch the graphs of ω, x, y, v_x, and v_y against φ.

(e) If each thread can withstand a maximum tension $T_{\mathrm{m}} = 10\,\mathrm{N}$, find the maximum height H_{m} to which we can wind the wheel such that the threads don't tear during the subsequent motion.

Data:
Acceleration due to gravity – $g = 9.81\,\mathrm{m/s^2}$

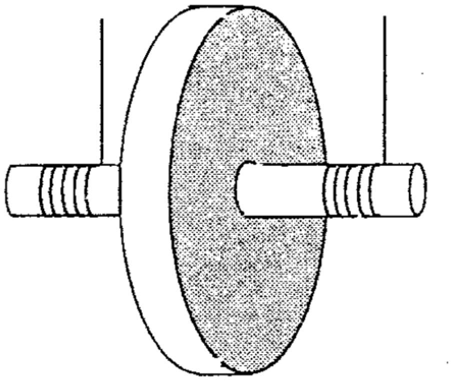

Figure 1

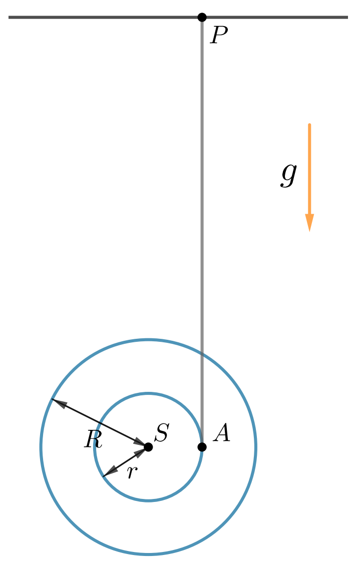

Figure 2

Solution

(a), (b), (c) We will solve the first three subparts in one pass. For a start, let's calculate the angular acceleration of the wheel ε. We'll consider the torque τ about point A as the wheel descends. There are only two vertical forces, the weight Mg at S and the net tension $2T$ at A. Thus $\tau = Mgr$. The moment of inertia of the wheel about A is $I_A = MR^2/2 + Mr^2$ by the parallel axis theorem. Now we can write $\tau = I_A\varepsilon$ because A is an instantaneous rotation axis. From there, we extract ε. Since there's no slipping at A, we can find the linear acceleration of the wheel a via $a = \varepsilon r$. The result is

$$a = \frac{2r^2}{R^2 + 2r^2} g. \tag{1}$$

After substituting this in the force equation $Mg - 2T = Ma$, we find the tension

$$\boxed{T = \frac{Mg}{2}\frac{R^2}{R^2 + 2r^2} = 1.95\,\mathrm{N}.} \tag{2}$$

Now we'll derive the dependence of the angular velocity ω on s. The linear velocity of the centre of the wheel is $v = \omega r$, so it suffices to find v. Denoting the time elapsed since the wheel was released by t, we have $v = at$ and $s = at^2/2$, or equivalently $v = \sqrt{2as}$. Hence $\omega = \sqrt{2as}/r$, or

$$\boxed{\omega = \sqrt{\frac{4gs}{R^2 + 2r^2}}.} \tag{3}$$

The translational kinetic energy is in turn $E_T = Mv^2/2 = Mas$, so

$$\boxed{E_T = \frac{2r^2}{R^2 + 2r^2} Mgs = 9.76 \times 10^{-3}\,\mathrm{J}.} \tag{4}$$

The rotational kinetic energy is $E_R = I\omega^2/2$, where the moment of inertia I about the axis of symmetry S is equal to $MR^2/2$. After simplifying,

$$E_R = \frac{R^2}{R^2 + 2r^2} Mgs. \tag{5}$$

And, assuming the gravitational potential energy is zero at the initial position of the wheel, its value is now $E_P = -Mgs$. Indeed, $E_T + E_R + E_P = 0$ from energy conservation. The ratios are

$$\boxed{\frac{E_T}{E_R} = \frac{2r^2}{R^2} = 5.00 \times 10^{-3}} \quad \text{and} \quad \boxed{\frac{E_T}{E_P} = -\frac{2r^2}{R^2 + 2r^2} = -4.98 \times 10^{-3}.} \tag{6}$$

(d) We'll introduce coordinates x and y with an origin at the position of A right as the wheel starts to turn around. The x-axis will point to the right and the y-axis will point downwards. Then, the coordinates of S at the beginning of the turn will be $(-r, 0)$. Throughout the turn, the problem involves four pieces of physics – forces along x, forces along y, torque, and the constraint that points A and P are always a distance L away. While this isn't conceptually difficult, the geometry is far from simple, and so the system of equations cannot be solved analytically.

It certainly helps that we're working in the special case of a long rope, $L \gg R \gg r$. The threads won't stay exactly vertical – denote their deviation by α. This corresponds to point A rising by $\Delta y_A = L(1 - \cos\alpha)$. The length scale of the zone covered by the wheel during the turn is on the order of r, so α is a small angle and hence $\Delta y_A \approx L\alpha^2/2$. If the x-displacement of point A is x_A, we have $\alpha \approx x_A/L$ and $\Delta y_A \approx x_A^2/2L$. Since x_A is of order r, we see that $\Delta y_A \ll r$. So, in our drawing on Figure 3, we constrain point A to a horizontal line.

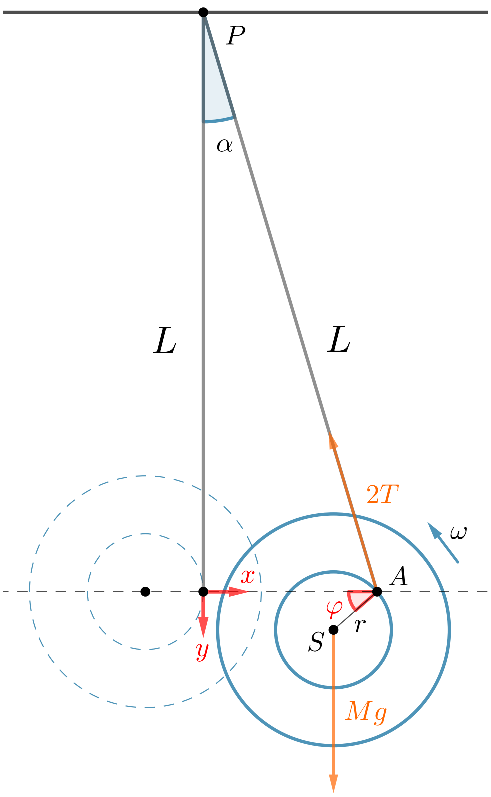

Figure 3

We're interested in the coordinates (x, y) of S, as well as the angular speed of the wheel ω. We write down the equations of motion for the wheel with the aim to express everything through the rotation angle φ:

$$Ma_y = Mg - 2T\cos\alpha \tag{7}$$

$$Ma_x = -2T\sin\alpha \tag{8}$$

$$2Tr\cos(\varphi - \alpha) = I\ddot{\varphi} \tag{9}$$

$$x = x_A - r\cos\varphi \tag{10}$$

$$y = r\sin\varphi \tag{11}$$

$$\alpha = \frac{x_A}{L} \tag{12}$$

These are not going to lead anywhere, so we try to simplify further. Differentiating (11) twice, we find

$$a_y = (r\cos\varphi)\ddot{\varphi} - (r\sin\varphi)\dot{\varphi}^2. \tag{13}$$

We neglect α in (7) and (9), and we substitute (9) and (13) into (7). We obtain the differential equation

$$\left(\frac{R^2}{r\cos\varphi} + r\cos\varphi\right)\ddot{\varphi} - (r\sin\varphi)\dot{\varphi}^2 - g = 0. \tag{14}$$

Tough luck – it's still too complicated. To wring out an answer for the angular velocity, we'll need to play fast and loose with the approximations. We'll make use of an argument from [1] and expand upon the discussion there.

Even though we can't directly obtain formulae for the physical quantities at hand, the equations above still provide us with some idea of their scale. For example, from (7) we know that a_y is of order g, and also that $2T$ is of the same order as Ma_y. Recall also that $x_A \sim r$. Using all this in (8), we conclude that a_x is of the same order as $a_y(r/L)$, meaning that $a_x \ll a_y$. Denoting the timescale of the wheel's turn by t', the typical displacements of S along x and y will be given by

$\Delta x \sim a_x t'^2$ and $\Delta y \sim a_y t'^2$. We can be certain that $\Delta y \sim r$ due to the geometry of the problem. Therefore $\Delta x \ll r$, so from now on we'll assume that S moves only along y (with point A still in motion along x). With hindsight, the diagram for the turn should look like this:

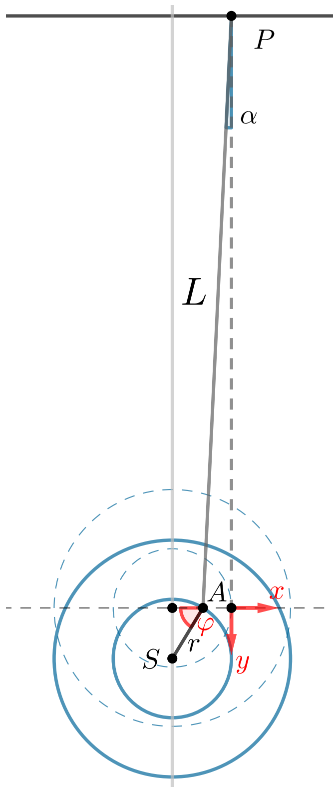

Figure 4

We can now claim

$$\boxed{x(\varphi) \approx -r,} \qquad \boxed{v_x(\varphi) \approx 0.} \tag{15}$$

It follows that the total speed of point S is $v = v_y = (r\cos\varphi)\dot{\varphi}$. We proceed to apply energy conservation. Set the potential energy to zero at the start of the turn. Back when the wheel was released, the potential energy was MgH and the kinetic energy was zero, so the total energy is always equal to MgH. For an arbitrary instant during the wheel's motion, the energy is

$$\frac{Mv^2}{2} + \frac{I\omega^2}{2} - Mgy = MgH. \tag{16}$$

Solving this with $I = MR^2/2$, $y = r\sin\varphi$, and $v = \omega r\cos\varphi$, we reach

$$\boxed{\omega(\varphi) \approx \sqrt{\frac{4g(H + r\sin\varphi)}{R^2 + 2r^2\cos^2\varphi}}.} \tag{17}$$

As a safety check, this formula indeed looks like (3) when $\varphi = 0$. Finally,

$$\boxed{y(\varphi) \approx -r\sin\varphi,} \qquad \boxed{v_y(\varphi) \approx \sqrt{\frac{4g(H + r\sin\varphi)}{(R/r)^2 + 2\cos^2\varphi}}\cos\varphi.} \tag{18}$$

Next, the graphs. Let's do some preliminary computations. For $\varphi = \{0, \pi\}$ the angular velocity is $\omega_0 = 104.14\,\mathrm{rad/s}$. The function $\omega(\varphi)$ clearly has a maximum at $\varphi = \pi/2$, because at that instant the numerator is maximised while the denominator is minimised. It's also evident that there aren't any other extrema. We evaluate $\omega_{\max} = 104.56\,\mathrm{rad/s}$. Because ω changes very little, the dominant effect in the graph of $v_y(\varphi)$ is the $\cos\varphi$ factor. So, it's essentially a sinusoid with an amplitude $v_0 = \omega_0 r = 0.31\,\mathrm{m/s}$. We can sketch the results as follows:

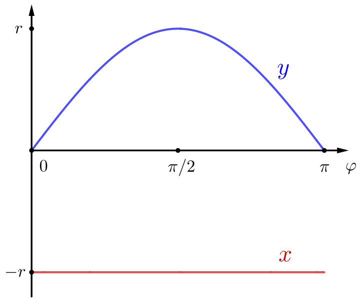

Figure 5

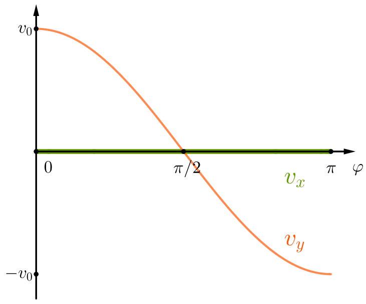

Figure 6

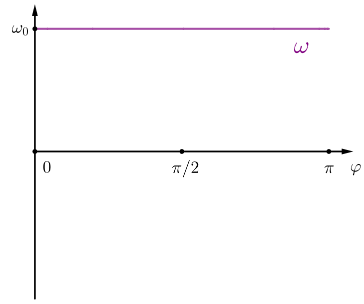

Figure 7

(e) We want the threads to hold even when the tension is maximal. We see from (7) that this happens when a_y is at a minimum. To get an expression for a_y, we ought to differentiate $v_y = \omega r \sin\varphi$ with respect to time. If we assume a constant ω, our work will become much easier, and the expression for a_y will still be very close to the truth. We find $a_y \approx -\omega^2 r \sin\varphi$, which is minimised for $\varphi = \pi/2$. There, $a_y \approx -\omega_{\mathrm{max}}^2 r$, and the tension at that instant is

$$T = \frac{1}{2}\left(Mg + M\omega_{\mathrm{max}}^2 r\right) = \frac{1}{2}Mg\left(1 + \frac{4(H+r)r}{R^2}\right). \tag{19}$$

The tension T will grow with H, and for $H = H_\mathrm{m}$ we have $T = T_\mathrm{m}$. Consequently,

$$\boxed{H_\mathrm{m} \approx \left(\frac{2T_\mathrm{m}}{Mg} - 1\right)\frac{R^2}{4r} - r = 1.23\,\mathrm{m}.} \tag{20}$$

All of this might seem dubious. It's prudent to verify with a simulation of (7)–(12) with the initial conditions $\omega = \omega_0$, $x = -r$, $y = 0$, $\dot{x} = 0$. We pick a large enough value for the length of the rope, $L = 4.0\,\mathrm{m}$. The results are presented on Figure 8. Note how tiny x and v_x remain, and also how close the values for y, v_y, and ω are compared to those from the analytical expressions above.

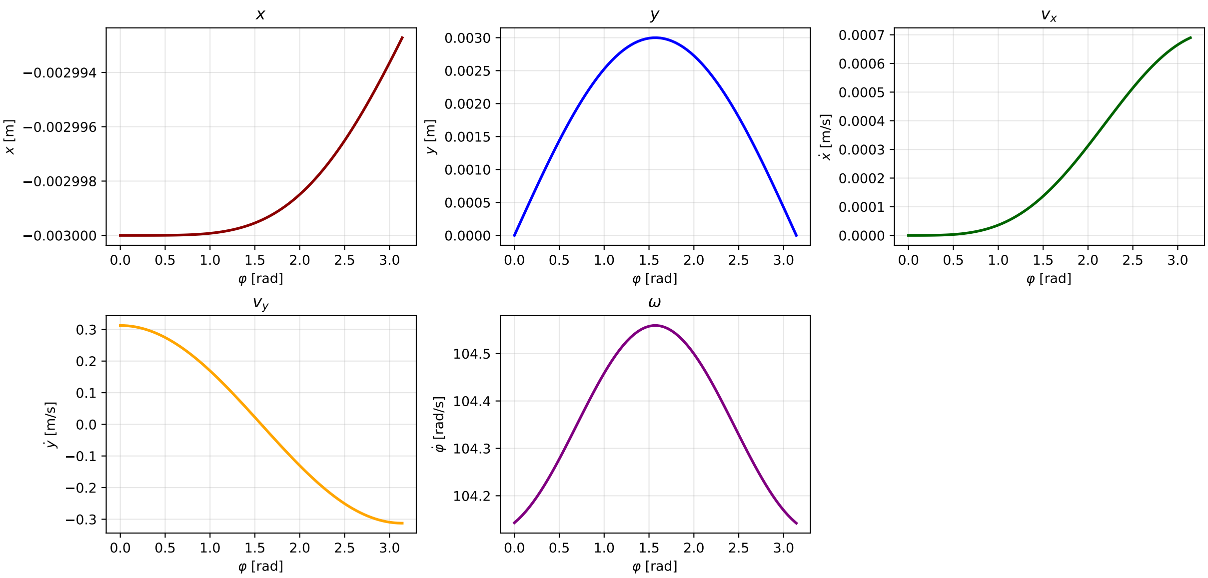

Figure 8

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References

[1] S. Kozel, V. Korovin, V. Orlov. *Sbornik zadach i zadaniy s otvetami i resheniyami* (Problem book with answers and solutions). Mnemozina, 2001, pp. 56-59.